# 《数据科学与工程数学基础》第4次作业答案

## 第一题

给定矩阵 $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$。

**解：**

特征值：$\lambda_1 = 1$，$\lambda_2 = 3$。

对应的单位特征向量：

$$\boldsymbol{q}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \boldsymbol{q}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

谱分解：

$$A = Q\Lambda Q^\top = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

## 第二题

给定矩阵 $A = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 3 & -1 \\ 0 & -1 & 3 \end{pmatrix}$。

**解：**

特征值：$\lambda_1 = 3 + \sqrt{2}$，$\lambda_2 = 3$，$\lambda_3 = 3 - \sqrt{2}$。

对应的单位特征向量：

$$\boldsymbol{q}_1 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}, \quad \boldsymbol{q}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \boldsymbol{q}_3 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}$$

谱分解：

$$A = Q\Lambda Q^\top = \begin{pmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 3+\sqrt{2} & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3-\sqrt{2} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{2}}{2} & \frac{1}{2} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{\sqrt{2}}{2} & \frac{1}{2} \end{pmatrix}$$

其中 $Q = [\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3]$，$\Lambda = \mathrm{diag}(3+\sqrt{2}, 3, 3-\sqrt{2})$。

## 第三题

给定矩阵 $C = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$。

**解：**

特征值：$\lambda_1 = 4$，$\lambda_2 = 1$（二重根）。

对应的单位特征向量：

$$\boldsymbol{q}_1 = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix}, \quad \boldsymbol{q}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\\0\end{pmatrix}, \quad \boldsymbol{q}_3 = \frac{1}{\sqrt{6}}\begin{pmatrix}1\\1\\-2\end{pmatrix}$$

谱分解：

$$C = Q\Lambda Q^\top = \begin{pmatrix}\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}}\\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}}\\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}}\end{pmatrix}\begin{pmatrix}4 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}}\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0\\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}}\end{pmatrix} = \begin{pmatrix}2 & 1 & 1\\1 & 2 & 1\\1 & 1 & 2\end{pmatrix}$$

逆矩阵：

$$C^{-1} = Q\Lambda^{-1}Q^\top = Q\begin{pmatrix}\frac{1}{4} & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{pmatrix}Q^\top$$

显式表达式（利用谱分解）：

$$C^{-1} = \frac{1}{4}\boldsymbol{q}_1\boldsymbol{q}_1^\top + \boldsymbol{q}_2\boldsymbol{q}_2^\top + \boldsymbol{q}_3\boldsymbol{q}_3^\top$$

计算得：

$$C^{-1} = \begin{pmatrix}\frac{3}{4} & -\frac{1}{4} & -\frac{1}{4}\\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4}\\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4}\end{pmatrix}$$

## 第四题

给定矩阵 $A = \begin{pmatrix}1 & 1\\2 & 0\\2 & 1\end{pmatrix}$。

**解：**

对第一列 $\boldsymbol{a}_1 = \begin{pmatrix}1\\2\\2\end{pmatrix}$ 构造Householder变换。

计算 $\|\boldsymbol{a}_1\| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3$。

选择 $\boldsymbol{e}_1 = \begin{pmatrix}1\\0\\0\end{pmatrix}$，令 $\boldsymbol{v} = \boldsymbol{a}_1 - \|\boldsymbol{a}_1\|\boldsymbol{e}_1 = \begin{pmatrix}1\\2\\2\end{pmatrix} - 3\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}-2\\2\\2\end{pmatrix}$。

计算 $\|\boldsymbol{v}\|^2 = (-2)^2 + 2^2 + 2^2 = 12$。

Householder矩阵：

$$H_1 = I - \frac{2}{\|\boldsymbol{v}\|^2}\boldsymbol{v}\boldsymbol{v}^\top = I - \frac{2}{12}\begin{pmatrix}-2\\2\\2\end{pmatrix}\begin{pmatrix}-2 & 2 & 2\end{pmatrix}$$

计算 $\boldsymbol{v}\boldsymbol{v}^\top$：

$$\boldsymbol{v}\boldsymbol{v}^\top = \begin{pmatrix}-2\\2\\2\end{pmatrix}\begin{pmatrix}-2 & 2 & 2\end{pmatrix} = \begin{pmatrix}4 & -4 & -4\\-4 & 4 & 4\\-4 & 4 & 4\end{pmatrix}$$

因此：

$$H_1 = \begin{pmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{pmatrix} - \frac{1}{6}\begin{pmatrix}4 & -4 & -4\\-4 & 4 & 4\\-4 & 4 & 4\end{pmatrix} = \begin{pmatrix}\frac{1}{3} & \frac{2}{3} & \frac{2}{3}\\\frac{2}{3} & \frac{1}{3} & -\frac{2}{3}\\\frac{2}{3} & -\frac{2}{3} & \frac{1}{3}\end{pmatrix}$$

应用Householder变换：

$$H_1A = \begin{pmatrix}\frac{1}{3} & \frac{2}{3} & \frac{2}{3}\\\frac{2}{3} & \frac{1}{3} & -\frac{2}{3}\\\frac{2}{3} & -\frac{2}{3} & \frac{1}{3}\end{pmatrix}\begin{pmatrix}1 & 1\\2 & 0\\2 & 1\end{pmatrix}$$

因此：

$$H_1A = \begin{pmatrix}3 & 1\\0 & 0\\0 & 1\end{pmatrix}$$

由于 $A$ 是 $3\times 2$ 矩阵，$R$ 应该是 $3\times 2$ 上三角矩阵。调整行顺序使得 $R$ 为上三角形式：

$$R = \begin{pmatrix}3 & 1\\0 & 1\\0 & 0\end{pmatrix}$$

由于 $H_1$ 是正交矩阵，且 $H_1A = \begin{pmatrix}3 & 1\\0 & 0\\0 & 1\end{pmatrix}$，我们需要调整 $Q$ 矩阵的行顺序以匹配 $R$。交换 $H_1$ 的第2、3行得到：

$$Q = \begin{pmatrix}\frac{1}{3} & \frac{2}{3} & \frac{2}{3}\\\frac{2}{3} & -\frac{2}{3} & \frac{1}{3}\\\frac{2}{3} & \frac{1}{3} & -\frac{2}{3}\end{pmatrix}$$

因此 $QR = A$，其中 $Q$ 是 $3\times 3$ 正交矩阵，$R$ 是 $3\times 2$ 上三角矩阵。

## 第五题

给定矩阵 $A = \begin{pmatrix} 0 & 3 & 1 \\ 0 & 4 & -2 \\ 2 & 1 & 2 \end{pmatrix}$。

**解：**

设 $A$ 的列向量为 $\boldsymbol{a}_1 = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$，$\boldsymbol{a}_2 = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix}$，$\boldsymbol{a}_3 = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$。

对第一列进行归一化：

$$\|\boldsymbol{a}_1\| = \sqrt{0^2 + 0^2 + 2^2} = 2$$

$$\boldsymbol{q}_1 = \frac{\boldsymbol{a}_1}{\|\boldsymbol{a}_1\|} = \frac{1}{2}\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad r_{11} = \|\boldsymbol{a}_1\| = 2$$

对第二列进行正交化。计算投影：$r_{12} = \boldsymbol{q}_1^\top \boldsymbol{a}_2 = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} = 1$。

正交化：

$$\boldsymbol{u}_2 = \boldsymbol{a}_2 - r_{12}\boldsymbol{q}_1 = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} - 1 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$$

归一化：

$$\|\boldsymbol{u}_2\| = \sqrt{3^2 + 4^2 + 0^2} = 5$$

$$\boldsymbol{q}_2 = \frac{\boldsymbol{u}_2}{\|\boldsymbol{u}_2\|} = \frac{1}{5}\begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{3}{5} \\ \frac{4}{5} \\ 0 \end{pmatrix}, \quad r_{22} = \|\boldsymbol{u}_2\| = 5$$

对第三列进行正交化。计算投影：

$$r_{13} = \boldsymbol{q}_1^\top \boldsymbol{a}_3 = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} = 2$$

$$r_{23} = \boldsymbol{q}_2^\top \boldsymbol{a}_3 = \begin{pmatrix} \frac{3}{5} & \frac{4}{5} & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} = \frac{3}{5} - \frac{8}{5} = -1$$

正交化：

$$\boldsymbol{u}_3 = \boldsymbol{a}_3 - r_{13}\boldsymbol{q}_1 - r_{23}\boldsymbol{q}_2 = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} - 2\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} - (-1)\begin{pmatrix} \frac{3}{5} \\ \frac{4}{5} \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + \begin{pmatrix} \frac{3}{5} \\ \frac{4}{5} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{8}{5} \\ -\frac{6}{5} \\ 0 \end{pmatrix}$$

归一化：

$$\|\boldsymbol{u}_3\| = \sqrt{\left(\frac{8}{5}\right)^2 + \left(-\frac{6}{5}\right)^2} = \sqrt{\frac{64+36}{25}} = \sqrt{\frac{100}{25}} = 2$$

$$\boldsymbol{q}_3 = \frac{\boldsymbol{u}_3}{\|\boldsymbol{u}_3\|} = \frac{1}{2}\begin{pmatrix} \frac{8}{5} \\ -\frac{6}{5} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{4}{5} \\ -\frac{3}{5} \\ 0 \end{pmatrix}, \quad r_{33} = \|\boldsymbol{u}_3\| = 2$$

因此：

$$Q = \begin{pmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 & \boldsymbol{q}_3 \end{pmatrix} = \begin{pmatrix} 0 & \frac{3}{5} & \frac{4}{5} \\ 0 & \frac{4}{5} & -\frac{3}{5} \\ 1 & 0 & 0 \end{pmatrix}$$

$$R = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{pmatrix} = \begin{pmatrix} 2 & 1 & 2 \\ 0 & 5 & -1 \\ 0 & 0 & 2 \end{pmatrix}$$

所以 $A = QR$。

## 第六题

**6.1** 给定矩阵 $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$。

**解：**

设 $A$ 的列向量为 $\boldsymbol{a}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$，$\boldsymbol{a}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$，$\boldsymbol{a}_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$。

对第一列进行归一化：

$$\|\boldsymbol{a}_1\| = \sqrt{0^2 + 1^2 + 1^2} = \sqrt{2}$$

$$\boldsymbol{q}_1 = \frac{\boldsymbol{a}_1}{\|\boldsymbol{a}_1\|} = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\1\end{pmatrix} = \begin{pmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\end{pmatrix}, \quad r_{11} = \|\boldsymbol{a}_1\| = \sqrt{2}$$

对第二列进行正交化。计算投影：$r_{12} = \boldsymbol{q}_1^\top \boldsymbol{a}_2 = \begin{pmatrix}0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}1\\1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}$。

正交化：

$$\boldsymbol{u}_2 = \boldsymbol{a}_2 - r_{12}\boldsymbol{q}_1 = \begin{pmatrix}1\\1\\0\end{pmatrix} - \frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\1\end{pmatrix} = \begin{pmatrix}1\\1\\0\end{pmatrix} - \begin{pmatrix}0\\\frac{1}{2}\\\frac{1}{2}\end{pmatrix} = \begin{pmatrix}1\\\frac{1}{2}\\-\frac{1}{2}\end{pmatrix}$$

归一化：

$$\|\boldsymbol{u}_2\| = \sqrt{1^2 + \left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2} = \sqrt{1 + \frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{3}{2}}$$

$$\boldsymbol{q}_2 = \frac{\boldsymbol{u}_2}{\|\boldsymbol{u}_2\|} = \frac{1}{\sqrt{\frac{3}{2}}}\begin{pmatrix}1\\\frac{1}{2}\\-\frac{1}{2}\end{pmatrix} = \sqrt{\frac{2}{3}}\begin{pmatrix}1\\\frac{1}{2}\\-\frac{1}{2}\end{pmatrix} = \begin{pmatrix}\sqrt{\frac{2}{3}}\\\sqrt{\frac{1}{6}}\\-\sqrt{\frac{1}{6}}\end{pmatrix}, \quad r_{22} = \|\boldsymbol{u}_2\| = \sqrt{\frac{3}{2}}$$

对第三列进行正交化。计算投影：

$$r_{13} = \boldsymbol{q}_1^\top \boldsymbol{a}_3 = \begin{pmatrix}0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix} = \frac{1}{\sqrt{2}}$$

$$r_{23} = \boldsymbol{q}_2^\top \boldsymbol{a}_3 = \begin{pmatrix}\sqrt{\frac{2}{3}} & \sqrt{\frac{1}{6}} & -\sqrt{\frac{1}{6}}\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix} = \sqrt{\frac{2}{3}} - \sqrt{\frac{1}{6}} = \sqrt{\frac{1}{6}}$$

正交化：

$$\boldsymbol{u}_3 = \boldsymbol{a}_3 - r_{13}\boldsymbol{q}_1 - r_{23}\boldsymbol{q}_2$$

$$= \begin{pmatrix}1\\0\\1\end{pmatrix} - \frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\1\end{pmatrix} - \sqrt{\frac{1}{6}}\cdot\sqrt{\frac{2}{3}}\begin{pmatrix}1\\\frac{1}{2}\\-\frac{1}{2}\end{pmatrix}$$
$$= \begin{pmatrix}1\\0\\1\end{pmatrix} - \begin{pmatrix}0\\\frac{1}{2}\\\frac{1}{2}\end{pmatrix} - \frac{1}{3}\begin{pmatrix}1\\\frac{1}{2}\\-\frac{1}{2}\end{pmatrix} = \begin{pmatrix}\frac{2}{3}\\-\frac{2}{3}\\\frac{2}{3}\end{pmatrix}$$

归一化：

$$\|\boldsymbol{u}_3\| = \sqrt{\left(\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2} = \sqrt{\frac{12}{9}} = \frac{2\sqrt{3}}{3}$$

$$\boldsymbol{q}_3 = \frac{\boldsymbol{u}_3}{\|\boldsymbol{u}_3\|} = \frac{3}{2\sqrt{3}}\begin{pmatrix} \frac{2}{3} \\ -\frac{2}{3} \\ \frac{2}{3} \end{pmatrix} = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}, \quad r_{33} = \|\boldsymbol{u}_3\| = \frac{2\sqrt{3}}{3}$$

因此：

$$Q = \begin{pmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 & \boldsymbol{q}_3 \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{\frac{2}{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \sqrt{\frac{1}{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\sqrt{\frac{1}{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$R = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{pmatrix} = \begin{pmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{\frac{3}{2}} & \sqrt{\frac{1}{6}} \\ 0 & 0 & \frac{2\sqrt{3}}{3} \end{pmatrix}$$

**6.2** 给定矩阵 $A = \begin{pmatrix} 1 & \frac{1}{2} & 5 \\ 1 & -\frac{1}{2} & 2 \\ -1 & \frac{1}{2} & -2 \\ 1 & -\frac{3}{2} & 0 \end{pmatrix}$。

**解：**

设 $A$ 的列向量为 $\boldsymbol{a}_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 1 \end{pmatrix}$，$\boldsymbol{a}_2 = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{3}{2} \end{pmatrix}$，$\boldsymbol{a}_3 = \begin{pmatrix} 5 \\ 2 \\ -2 \\ 0 \end{pmatrix}$。

对第一列进行归一化：

$$\|\boldsymbol{a}_1\| = \sqrt{1^2 + 1^2 + (-1)^2 + 1^2} = \sqrt{4} = 2$$

$$\boldsymbol{q}_1 = \frac{\boldsymbol{a}_1}{\|\boldsymbol{a}_1\|} = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix}, \quad r_{11} = \|\boldsymbol{a}_1\| = 2$$

对第二列进行正交化。计算投影：$r_{12} = \boldsymbol{q}_1^\top \boldsymbol{a}_2 = \frac{1}{2}\begin{pmatrix} 1 & 1 & -1 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{3}{2} \end{pmatrix} = \frac{1}{2}\left(\frac{1}{2} - \frac{1}{2} - \frac{1}{2} - \frac{3}{2}\right) = -1$。

正交化：

$$\boldsymbol{u}_2 = \boldsymbol{a}_2 - r_{12}\boldsymbol{q}_1 = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{3}{2} \end{pmatrix} - (-1) \cdot \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{3}{2} \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

归一化：

$$\|\boldsymbol{u}_2\| = \sqrt{1^2 + 0^2 + 0^2 + (-1)^2} = \sqrt{2}$$

$$\boldsymbol{q}_2 = \frac{\boldsymbol{u}_2}{\|\boldsymbol{u}_2\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ -\frac{1}{\sqrt{2}} \end{pmatrix}, \quad r_{22} = \|\boldsymbol{u}_2\| = \sqrt{2}$$

对第三列进行正交化。计算投影：

$$r_{13} = \boldsymbol{q}_1^\top \boldsymbol{a}_3 = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \\ -2 \\ 0 \end{pmatrix} = \frac{1}{2}(5 + 2 + 2 + 0) = \frac{9}{2}$$

$$r_{23} = \boldsymbol{q}_2^\top \boldsymbol{a}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \\ -2 \\ 0 \end{pmatrix} = \frac{5}{\sqrt{2}}$$

正交化：

$$\boldsymbol{u}_3 = \boldsymbol{a}_3 - r_{13}\boldsymbol{q}_1 - r_{23}\boldsymbol{q}_2$$

$$= \begin{pmatrix} 5 \\ 2 \\ -2 \\ 0 \end{pmatrix} - \frac{9}{2} \cdot \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ 1 \end{pmatrix} - \frac{5}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 \\ 2 \\ -2 \\ 0 \end{pmatrix} - \begin{pmatrix} \frac{9}{4} \\ \frac{9}{4} \\ -\frac{9}{4} \\ \frac{9}{4} \end{pmatrix} - \begin{pmatrix} \frac{5}{2} \\ 0 \\ 0 \\ -\frac{5}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{4} \\ -\frac{1}{4} \\ -\frac{1}{4} \\ \frac{1}{4} \end{pmatrix}$$

归一化：

$$\|\boldsymbol{u}_3\| = \sqrt{\left(\frac{1}{4}\right)^2 + \left(-\frac{1}{4}\right)^2 + \left(-\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^2} = \sqrt{\frac{4}{16}} = \frac{1}{2}$$

$$\boldsymbol{q}_3 = \frac{\boldsymbol{u}_3}{\|\boldsymbol{u}_3\|} = 2\begin{pmatrix} \frac{1}{4} \\ -\frac{1}{4} \\ -\frac{1}{4} \\ \frac{1}{4} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix}, \quad r_{33} = \|\boldsymbol{u}_3\| = \frac{1}{2}$$

因此：

$$Q = \begin{pmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 & \boldsymbol{q}_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} \end{pmatrix}$$

$$R = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{pmatrix} = \begin{pmatrix} 2 & -1 & \frac{9}{2} \\ 0 & \sqrt{2} & \frac{5}{\sqrt{2}} \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$